```
/**
 * 🧩 Inventory Manager with Generics
 */

data class Item(
    val name: String,
    val category: String,
    val quantity: Int
)

fun printInventory(items: List<Item>) {
    if (items.isEmpty()) {
        println("No items found.")
        return
    }

    for (item in items) {
        println("Name: ${item.name} | Category: ${item.category} | Qty: ${item.quantity}")
    }
}

fun <T> filterList(list: List<T>, condition: (T) -> Boolean): List<T> {
    val result = mutableListOf<T>()
    for (element in list) {
        if (condition(element)) {
            result.add(element)
        }
    }
    return result
}

fun main() {
```

```
val inventory = listOf(
    Item("Laptop", "Electronics", 12),
    Item("Keyboard", "Electronics", 5),
    Item("Chair", "Furniture", 20),
    Item("Table", "Furniture", 8),
    Item("Notebook", "Stationery", 50),
    Item("Pen", "Stationery", 6)
)

println("\n--- Full Inventory ---")
printInventory(inventory)

val lowStock = filterList(inventory) { it.quantity < 10 }
println("\n--- Low Stock (<10) ---")
printInventory(lowStock)

val electronicsItems = filterList(inventory) { it.category == "Electronics" }
println("\n--- Electronics Category ---")
printInventory(electronicsItems)

println("\n--- Total Quantity per Category ---")

val categoryTotals = inventory
```